CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M. Sc. (BT/MI/BC) Sem – I Examination (Backlog), 2011-12

BT/MI/BC - 704 Molecular Biology

Date: 19.04.12, Thursday **Time: 10:00 a.m. to 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A **Total marks: 20**

Q1 Choose the correct option and put √ mark in front of it:

1 The proofreading of newly synthesized DNA is done by:
(a) Restriction endonuclease (b) DNA gyrase
(c) DNA ligase (d) DNA polymerase III

2 *E. coli* chromosomes
(a) Replicate in a unidirectional pattern from a single, unique origin, called *oriC*.
(b) Adopt a θ-shaped conformation during replication.
(c) Consist of two Y-shaped replication forks which move in similar directions
(d) Have AT rich regions at the origin of replication to maintain a tight conformation

3 Arthur Kornberg is attributed with the first *in vitro* synthesis of DNA. This required:
(a) A primer DNA, providing a free 5′ terminus to add nucleotides to.
(b) A template DNA which directs the sequence of the newly synthesized strand.
(c) The nucleotides A, T, G, and U.
(d) Mn^{2+} ions.

4 The following statements about telomerase are true:
(a) It extends the 3' end of a linear chromosome one repeat at a time.
(b) It uses a built-in DNA to act as a template for synthesis of the telomeric repeat.
(c) Without telomerase, linear chromosomes would become progressively shorter.
(d) a and c.

5 A striking feature of the genetic code is that an amino acid may be specified by more than one codon. This trait of genetic code is termed as ___________

6 The stage of translation in which an aminoacyl-tRNA binds to AUG on the P site of the ribosome is
(a) Initiation (b) Elongation (c) Termination (d) None of the above

P.T.O.

7 A synthetic mRNA of repeating sequence 5'-CACACACACACACACAC... is used for a cell-free protein synthesizing system. If we assume that protein synthesis can begin without the need for an initiator codon, what product/s are expected to occur after protein synthesis?

(a) One protein, consisting of a single amino acid
(b) Two proteins, each with an alternating sequence of two different amino acids
(c) One protein, with an alternating sequence of three different amino acids
(d) One protein, with an alternating sequence of two different amino acids

8 The amino acid arm of t-RNA has the following sequence at 3' end

(a) CCA (b) CCC (c) CAA (d) CAC

9 RNA polymerase II transcribes

(a) mRNA (b) rRNA (c) tRNA (d) All the above

10 An RNA with catalytic activity is known as ribozyme

☐True ☐ False

11 Transcription takes place in

(a) Cytoplasm (b) Nucleus (c) Golgi Complex (d) Endoplasmic reticulum

12 RNA editing involves use of

(a) gRNA (b) rRNA (c) tRNA (d) miRNA

13 A region of DNA where repressor protein can bind and inhibit transcription is called ____________.

14 The product of lacI is

(a) Monomeric repressor (b) Dimeric repressor
(c) Tetrameric repressor (d) Tetrameric inducer

15 Which of the following can physically block interactions between promoter DNA sequences and the proteins needed to initiate transcription?

(a) RNA (b) Nucleosome (c) Histone proteins (d) HO gene

16 High concentration of glucose reduces CRP activity

☐ True ☐ False

17 Gene density in the genomes of yeast, *Drosophila* and human is in the following order

(a) yeast>*Drosophila*>human (b) *Drosophila*>human>yeast
(c) human>*Drosophila*>yeast (d) yeast= *Drosophila*=human

18 DNA is said to be negatively supercoiled when ΔLk us

(a) < zero (b) > zero (c) = zero (d) =1

19 Telomeric DNA is an example of

(a) Minisatellites (b) Microsatellite (c) Pseudogene (d) Gene fragment

20 During reassociation of DNA, a point in the reaction when one half of the DNA is present as double stranded fragments is called as

(a) T_m (b) $C_ot_{1/2}$ (c) Linking number (d) Superhelical density

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Part B **Total Marks: 50**

Section I

Q1 (A) Answer any one of the following: **(04)**

(a) Giving labeled diagram describe the structure of nucleosome.

(b) What are topoisomerases? Explain the mechanism of topoisomerase I action.

(c) Describe the structure of eukaryotic mRNA

Q1 (A) Write short note on any two of the following: **(06)**

(a) Satellite RNAs

(b) Gene families

(c) Supercoiling in DNA

(d) DNA curvature

Q2 (B) Write short note on any three **(06)**

(a) Eukaryotic mRNA capping.

(b) Transcription Factors

(c) Bacterial RNA polymerase

(d) Rho-dependent termination

Q2 (B) Write a short note on any two of the following: **(04)**

(a) 5' and 3' modifications of mRNA

(b) CTD

(c) Transcription unit

Section II

Q.3 Attempt any two **(10)**

(a) Describe how DNA replication is initiated at an origin of replication to produce two replication forces that proceed bidirectionally away from each other.
(b) Compare eukaryotic replication versus prokaryotic replication.
(c) Describe in detail the steps involved with Excision Repair

Q.4 (A) Answer the following: (Any two) **(04)**

(a) Draw a neatly labeled structure of f-met-t-RNA
(b) Describe any **four** post-translational modifications.
(c) Distinguish between translation initiation of prokaryotes and eukaryotes.

Q.4 (B) Answer the following (Any two) **(06)**

(a) Describe the salient features of the genetic code with special emphasis on wobble hypothesis.
(b) Aminoacylation of t-RNA by aminoacyl t-RNA synthetase I and aminoacyl t-RNA synthetase II
(c) Describe in detail the process of translation in prokaryotes.

Q.5 (A) Answer in detail (Any three) **(09)**

(a) Describe the structure and role of lac repressor protein in regulation of lac operon.
(b) Describe the significance of guanosine pentaphosphate in stringent control.
(c) Describe the regulation of lamda genome for lysogenic pathway.
(d) Describe in detail the regulation mechanism of *trp* operon.

Q.5 (B) Define any one **(01)**

(a) RNA interference
(b) Gratitous inducer